

Design and Implementation of a Database Management System for Research Project, Publication and Grant Management

DBMS Course Project

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Problem Statement and Objectives

Problem Statement

Research-related information such as researchers, projects, publications and grants can become difficult to manage when stored manually or separately.

- Data duplication
- Inconsistent information
- Difficulty finding related records
- Difficult tracking of research projects
- Difficult tracking of publications and funding
- Lack of a centralized database system

Objectives

- Store researcher information in a structured database.
- Manage research projects and their details.
- Manage publications related to researchers and projects.
- Manage grants and funding information.
- Connect related data using primary keys and foreign keys.
- Retrieve and manage information using SQL.
- Provide a simple user interface using Streamlit.

System Workflow

- 01 Problem Identification
- 02 Requirements Analysis
- 03 ER Diagram / Database Design
- 04 Relational Schema
- 05 Normalization
- 06 MySQL Database Implementation
- 07 SQL Operations
- 08 Python Database Connectivity
- 09 Streamlit Application
- 10 Dashboard / Data Management
- 11 Results

Database Design and Relational Schema

Database: ResearchManagement. There are exactly four tables.

Researcher

- ResearcherID (PK)
- Name
- Email
- Department

Project

- ProjectID (PK)
- ProjectName
- StartDate
- EndDate
- ResearcherID (FK)

Publication

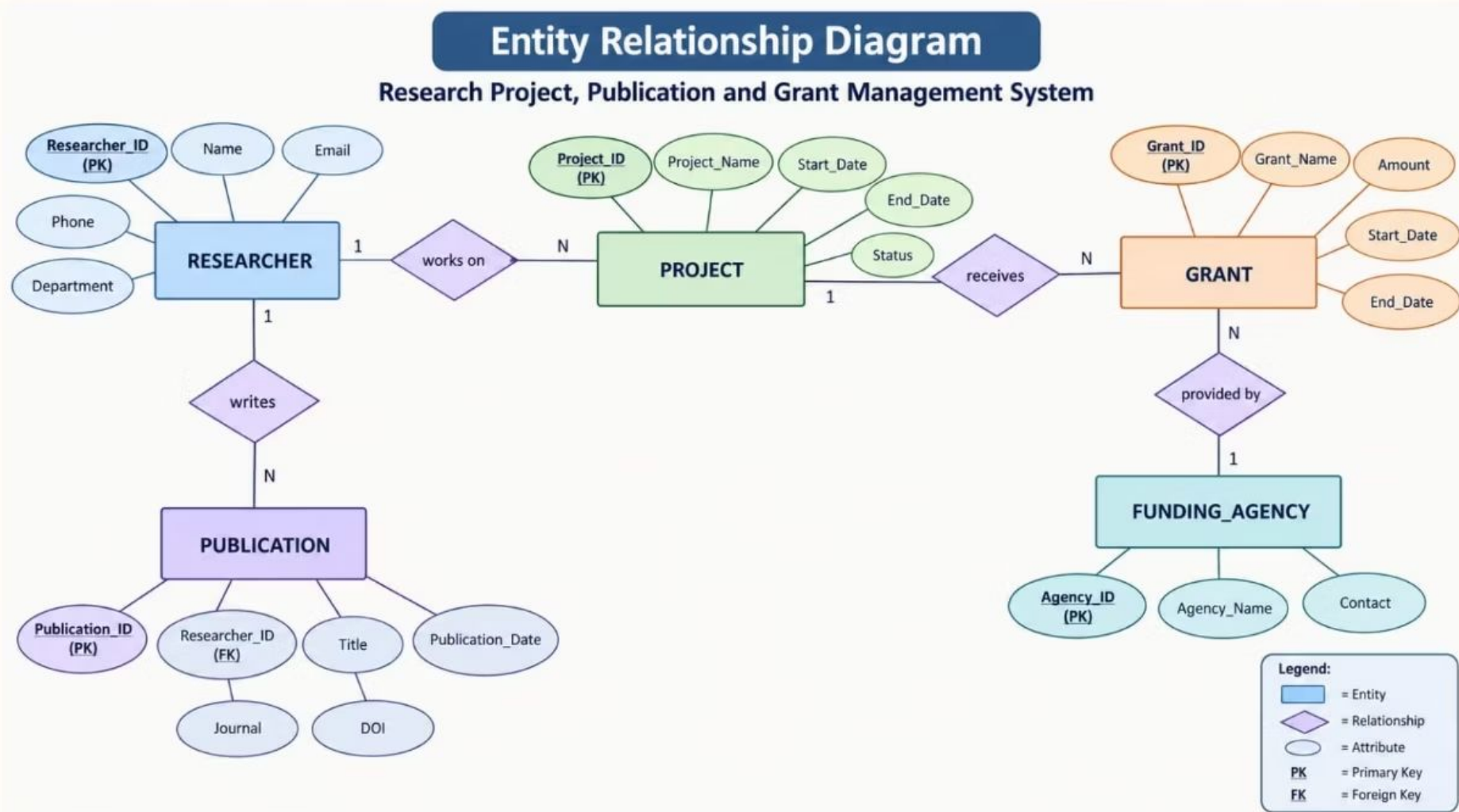
- PublicationID (PK)
- Title
- Journal
- PublicationYear
- ResearcherID (FK)
- ProjectID (FK)

Grant

- GrantID (PK)
- GrantName
- Amount
- FundingAgency
- ProjectID (FK)

Entity Relationship Diagram

SYSTEM DESIGN



Database Normalization

1NF — First Normal Form

- Each field contains a single atomic value.
- No repeating groups are stored in a single field.

2NF — Second Normal Form

- No partial dependency.
- Partial dependency occurs when a non-key attribute depends only on part of a composite primary key.
- Our tables use single-column primary keys, so partial dependency does not occur.

3NF — Third Normal Form

- No unnecessary transitive dependency.
- Independent entities are stored in separate tables: Researcher, Project, Publication and Grant.
- Foreign keys connect related information instead of unnecessarily duplicating entity details.

SQL Operations and Database Implementation

- CREATE → Creates the database and tables.
- INSERT → Adds records.
- SELECT → Retrieves records.
- UPDATE → Modifies existing records.
- DELETE → Removes records.
- JOIN → Combines related information from multiple tables.
- SUM / GROUP BY → Used for summary and funding calculations.

```
SELECT SUM(Amount) FROM `Grant`;
```

Result: \$1,005,000

Database Implementation

- Primary Keys uniquely identify each record.
- Foreign Keys connect related tables and maintain relationships.
- Constraints help maintain data integrity.

Application Architecture

1. USER
2. STREAMLIT USER INTERFACE
3. PYTHON APPLICATION
4. DATABASE CONNECTION
5. MYSQL
6. ResearchManagement DATABASE
7. Researcher / Project / Publication / Grant TABLES

- MySQL stores the project database.
- Python handles database operations and application logic.
- mysql-connector-python connects Python with MySQL.
- Streamlit provides the web-based user interface.
- Pandas helps display retrieved database records in tables.

Main Project Files

- app.py → Streamlit interface and dashboard
- database.py → MySQL connection and database operations
- schema.sql → Database and table creation
- sample_data.sql → Sample database records
- requirements.txt → Required Python packages

System Demonstration and Results

Actual Results

Researchers: 4

Projects: 4

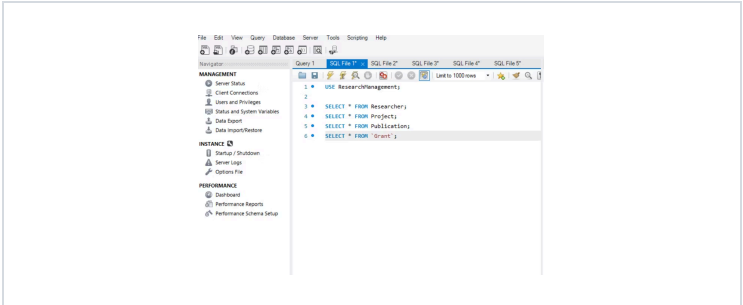
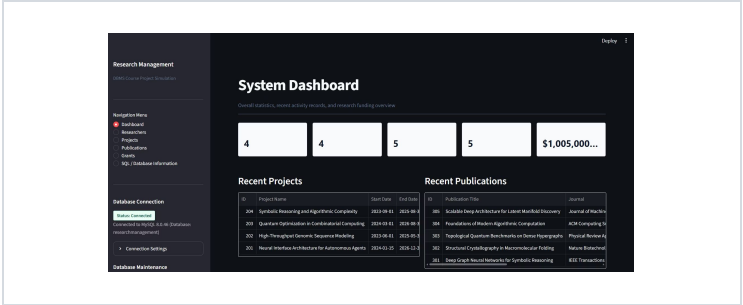
Publications: 5

Grants: 5

Total Funding: \$1,005,000

Dashboard Functions

- Dashboard summary
- Researchers
- Projects
- Publications
- Grants
- Recent Projects
- Recent Publications
- Database records retrieved from MySQL



System flow: User → Streamlit Dashboard → Python → MySQL → ResearchManagement Database → Data displayed in Dashboard.

Conclusion and Future Scope

Conclusion

- A centralized relational database was successfully designed.
- Four main tables were created: Researcher, Project, Publication and Grant.
- Primary keys uniquely identify records.
- Foreign keys connect related tables.
- Normalization reduces redundancy and improves consistency.
- SQL operations are used to manage and retrieve data.
- MySQL was successfully connected with Python.
- Streamlit provides the user interface.
- The dashboard successfully retrieves and displays database information.

Future Scope

- Add user authentication.
- Add advanced search and filtering.
- Add more reports and charts.
- Add more research management features.
- Deploy the application online.

GitHub:

https://github.com/Sharvari316/DBMS_COURSE_PROJECT

Thank You